

Announcements

Today

- Binomial distribution as emergent process
 - agent based model
- Using modern RNG libraries
- Events in continuous time
- Exponential distribution

Agent-based model

- Another outcome: **number dead** out of 100 individuals

Algorithm

set each individual to “alive”

set each individual to $p_death = 0.2$

for each individual

 generate U from $U(0,1)$

 if $U \leq p_death$

 set individual to “dead”

count dead

Binomial distribution

- **Emerges** from the agent-based simulation
- Underlying process: stochastic binary events
- Number of events that occur out of N possible
- **Higher-level abstraction** of the biological process or ABM is possible
 - i.e. simplifies to: $N_{\text{dead}} \sim \text{Binomial}(N, p)$

Binomial distribution

- Sum of independent stochastic binary events with the same probability

- $\Pr(X = x \mid n, p) =$

Discrete x
 $0, 1, 2, \dots, n$

$$f(x, n, p) = \binom{n}{x} p^x (1-p)^{n-x}$$

Probability
mass
function
(pmf)

$$X \sim \text{Binomial}(n, p)$$

Binomial coefficient:
how many ways to
choose x from n

“Probability that x events occur out of n possible”

Modern algorithms

1. generate random bits
2. turn those into integers
3. divide by largest integer to give $\text{Uniform}(0,1)$

Modern algorithms

- **R:** Mersenne_Twister
- **Numpy:** PCG64 (Permuted Congruential Generator)

Discrete vs continuous time

- Discrete: all events in a period
- Continuous: each event occurs at a particular time

Events in continuous time

- When does the event occur?
 - “time to event”
 - “waiting time”

When does the individual die?

- Approximation
 - small interval of time Δt
 - constant probability of death, $\Pr(\text{death})$
- Diagrams
 - Event sequence diagram
 - Flowchart

When does the individual die?

Algorithm

set c , Δt

set state = alive, $t = 0$

while state = alive

$t = t + \Delta t$

 generate U from $U(0,1)$

 if $U \leq p_{\text{death}}$

 state = dead

print t , time to death